

Grade 6 Accelerated
Unit 12
Lesson 1 Homework

Name Answer Key
Date _____
Period _____

A map is set to a scale factor of $32 \text{ mm} = 1 \text{ km}$. Use this scale to answer questions 1-3.

1. Two mountains are 451.1 mm apart on the map. Create a proportion that could be used to determine the actual distance between the mountains.

$$\frac{32 \text{ mm}}{1 \text{ km}} = \frac{451.1 \text{ mm}}{x \text{ km}}$$

2. Use proportions to solve for the actual distance between the mountains. Round the answer to the nearest tenth.

$$32 \cdot x = 1 \cdot 451.1$$

$$x = 14.1 \text{ km}$$

$$\frac{32x}{32} = \frac{451.1}{32}$$

$$x = 14.096875$$

3. Using the scale from the map in question 1, if the distance between two towns is 25 kilometers, how many millimeters apart will the towns be on the map?

$$\frac{32 \text{ mm}}{1 \text{ km}} = \frac{x \text{ mm}}{25 \text{ km}}$$

$$32 \cdot 25 = 1x$$

$$800 = x$$

The towns will be 800 mm apart on the map.

In a construction zone, there are black and gray beams that are being used for a building that are proportional in size. The black beams are 5 feet wide and 16 feet long, and the gray beams are 12.5 ft long. Use this information to answer questions 4-5.

4. Fill in the remaining values for the proportion below.

$$\frac{5}{x} = \frac{16}{12.5}$$

5. Using the above proportion, determine the width of the gray beams.

$$5 \cdot 12.5 = 16x$$

The gray beam are 3.9 feet wide.

$$\frac{62.5}{16} = \frac{16x}{16}$$

$$3.90625 = x$$

Use the following information to solve problems 6-10. Suppose 8,800 people live in apartment buildings in Charlotte County, Florida. A sample of 310 people were polled asking them, if given the opportunity, would they move into a new house.

- 144 said yes.
- 150 said no.
- 16 were undecided.

6. Create a proportion to determine the total number of people who were undecided.

$$\frac{16 \text{ undecided}}{310 \text{ polled people}} = \frac{x \text{ undecided}}{8,800 \text{ people in city}}$$

$$16 \cdot 8,800 = 310x$$

$$\frac{140,800}{310} = \frac{310x}{310} \approx 454 \text{ people were undecided}$$

$$454.193584 = x$$

$$\frac{16}{310} = \frac{x}{8,800}$$

7. Create a proportion to determine the total number of people who said yes.

$$\frac{144 \text{ yes}}{310 \text{ polled}} = \frac{x \text{ yes}}{8,800 \text{ people}}$$

$$= 144 \cdot 8,800 = 310x$$

$$\frac{1,267,200}{310} = \frac{310x}{310} \approx 4,088 \text{ people said yes}$$

$$4,087.7419 = x$$

$$\frac{144}{310} = \frac{x}{8,800}$$

8. Create a proportion to determine the total number of people who said no.

$$\frac{150 \text{ no}}{310 \text{ polled}} = \frac{x \text{ no}}{8,800 \text{ people}}$$

$$= 150 \cdot 8,800 = 310x$$

$$\frac{1,320,000}{310} = \frac{310x}{310} \approx 4,258 \text{ people said no}$$

$$4,258.06452x$$

$$\frac{150}{310} = \frac{x}{8,800}$$

9. To the nearest whole number, determine how many people were undecided.

$$16 \cdot 8,800 = 310x$$

$$\frac{140,800}{310} = \frac{310x}{310} \approx 454 \text{ people were undecided}$$

$$x = 454.193584$$

10. To the nearest whole number, determine the number of people who said yes.

$$144 \cdot 8,800 = 310x$$

$$\frac{1,267,200}{310} = \frac{310x}{310} \approx 4,088 \text{ people said yes}$$

$$4,087.7419 = x$$

Grade 6 Accelerated
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Lesson 2 Homework

Name Answer Key
Date _____
Period _____

Solve each problem using proportional reasoning. Show your work.

1. Sashia, a competitive swimmer, swims at a constant speed of 50 meters in 56 seconds. To the nearest tenth of a second, how long does it take her to swim 228 meters?

$$x = 255.36$$

$$x = 255.34 \text{ seconds}$$

$$\frac{50}{56} = \frac{228}{x}$$

2. To make a shade of paint called international orange, mix 3 quarts of yellow paint with 24 cups of red paint. How much red paint should be mixed with 4.5 cups of yellow paint to make international orange?

$$\frac{12x}{12} = \frac{108}{12}$$

$$x = 9$$

9 cups of red paint are needed

3 quarts = 12 cups

$$\frac{12}{24} = \frac{4.5}{x}$$

3. It takes an ant farm 4 days to consume $\frac{1}{6}$ of an apple. At that rate, in how many days will the ant farm consume 9 apples?

$$x = 216$$

$$\frac{\frac{1}{6}}{4} = \frac{9}{x}$$

In 216 days the ants will consume 9 apples.

4. Raspberries cost \$3.23 per $\frac{1}{2}$ pint. How much would it cost to purchase 3 pints of blackberries?

$$x = 19.38$$

It would cost \$19.38 to purchase 3 pints of raspberries.

$$\frac{0.5}{3.23} = \frac{3}{x}$$

5. Thiago is going on a road trip with his family. He wants to find out how long he will be in the car. His mother told him that their destination is 2,140 miles away and she will be driving at a constant 60 miles per hour. To the nearest tenth of an hour, how long will it take Thiago and his family to reach their destination?

$$x = 35.\bar{6}$$

It will take Thiago and his family 35.7 hours to reach their destination.

$$\frac{60}{1} = \frac{2140}{x}$$

6. An 18-inch pizza from Pizza Pi is cut into 12 slices and costs \$9.95. Rita has enough pizza for each person to eat 2 slices. If Rita has \$29.85 worth of pizza, how many people can she feed?

$$\frac{9.95}{12} = \frac{29.85}{x}$$

$$x = 36$$

Divide by 2 slices per person, so 18 people can eat.

Rita can feed 18 people.

7. Cashews cost \$63.00 per 5 pound bag. How much would a 2 pound bag of cashews cost?

$$\frac{63}{5} = \frac{x}{2}$$

$$126 = 5x$$

$$x = 25.2$$

A 2 pound bag of cashews would cost \$25.20.

8. It takes Richard 30 minutes to mow $\frac{1}{4}$ of an acre of grass. If his yard is 1.6 acres, how many hours will it take him to mow the lawn?

$$\frac{30}{.25} = \frac{x}{1.6}$$

$$48 = .25x$$

$$x = 192 \text{ min.}$$

$$\frac{192}{60} = 3.2$$

It will take Richard 3.2 hours to mow the lawn.

9. Randi wants to paint the rooms in her house. It takes her 24 hours to paint all 4 rooms with 1 coat of paint each. How many rooms can she paint each day if she can spend 8 hours a day painting?

$$\frac{24}{4} = \frac{8}{x}$$

$$24x = 32$$

$$x = 1\frac{1}{3}$$

She can paint $1\frac{1}{3}$ rooms a day.

10. Typing at a rate of 40 words per minute is considered a good speed. If a student has to type a paper that contains 700 words, how many minutes will it take the student to type the paper?

$$\frac{40}{1} = \frac{700}{x}$$

$$40x = 700$$

$$x = 17.5$$

It will take the student 17.5 minutes to type the paper.

Solve each problem using proportional reasoning. Show your work.

For questions 1-2, use the following scenario:

A recipe for salad dressing calls for 3 parts oil for every 1 part vinegar.

1. How many tablespoons (tbsp) of vinegar are needed to make $\frac{3}{4}$ cup of salad dressing? 1 cup = 16 tbsp

$$\frac{1 \text{ cup}}{16 \text{ tbsp}} = \frac{\frac{3}{4} \text{ cup}}{12 \text{ tbsp}} \quad \text{oil : vinegar : total} \rightarrow 3 : 1 : 4$$

$$\frac{\text{vinegar}}{\text{total}} \left| \frac{1}{4} = \frac{x}{12 \text{ tbsp}} \right. \quad x = 3 \text{ tbsp of vinegar}$$

2. How many tablespoons of oil are needed to make $\frac{3}{4}$ cup of salad dressing?

$$\frac{\text{oil}}{\text{total}} \left| \frac{3}{4} = \frac{x}{12 \text{ tbsp}} \right. \quad x = 9 \text{ tbsp of oil}$$

3. The retail price of grapes is \$2.10 per pound. If a customer paid \$7.35 for grapes, how many pounds did she buy?

$$\frac{\$}{\text{lb}} \left| \frac{2.10}{1} = \frac{7.35}{x} \right. \quad x = 3.5 \text{ pounds}$$

4. Gregg and Dan are moving boxes. Gregg can move 8 boxes every half hour. Dan can move 11 boxes every half hour. How many hours will it take Gregg and Dan to move all 114 boxes?

$$\frac{\text{boxes}}{\text{hours}} \left| \frac{19}{0.5} = \frac{114}{x} \right. \quad x = 3 \text{ hours}$$

For questions 5-7, use the following scenario:

A recipe for sparkling cranberry pineapple punch calls for 3 cups of pineapple juice, 3 cups of cranberry juice, and 1 quart of ginger ale. Miguel uses to make 7.5 cups of pineapple juice to make punch for his party.

5. How much ginger ale does Miguel need to use for his punch?

$$\begin{array}{c|c} \frac{\text{pineapple}}{\text{ginger ale}} & \frac{3 \text{ cups}}{1 \text{ qt}} = \frac{7.5 \text{ cups}}{x} \end{array} \quad x = 2.5 \text{ quarts}$$

$\xrightarrow{\times 2.5}$
 $\xleftarrow{\times 2.5}$

6. How much cranberry juice does Miguel need to use for his punch?

$$\begin{array}{c|c} \frac{\text{cran}}{\text{pine}} & \frac{3 \text{ cups}}{3 \text{ cups}} = \frac{x}{7.5 \text{ cups}} \end{array} \quad x = 7.5 \text{ cups}$$

$\xrightarrow{\times 2.5}$
 $\xleftarrow{\times 2.5}$

7. By what factor did Miguel scale the recipe up?

2.5 He made 2.5 times the original recipe.

For questions 8-10, use the following scenario:

A teacher is planning a field trip to a theme park. For every 8 students, the teacher must have 1 chaperone.

8. What is the ratio of students to total people on the field trip?

8:9

9. If the teacher purchased 144 total tickets, how many tickets were for students?

$$\begin{array}{c|c} \frac{\text{students}}{\text{total}} & \frac{8}{9} = \frac{x}{144} \end{array} \quad x = 128 \text{ student tickets}$$

$\xrightarrow{\times 16}$
 $\xleftarrow{\times 16}$

10. If the teacher purchased 108 total tickets, how many tickets were for chaperones?

$$\begin{array}{c|c} \frac{\text{chaperones}}{\text{total}} & \frac{1}{9} = \frac{x}{108} \end{array} \quad x = 12 \text{ chaperone tickets}$$

$\xrightarrow{\times 12}$
 $\xleftarrow{\times 12}$

Grade 6 Accelerated
Unit 12
Lesson 4 Homework

Name Answer Key
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Period _____

1. A store is having a storewide $\frac{1}{4}$ off sale. What is the amount saved on an item that costs \$7.25?

$$0.25 \cdot 7.25 = \$1.81$$

2. John bought a dented washing machine because it was discounted $\frac{3}{20}$ off the original price of \$650. What was the reduced price, in dollars?

$$\frac{3}{20} \cdot 650 = \$97.50 \text{ off}$$

$$\begin{array}{r} 6 \quad 5 \quad 0 \quad .0 \quad 0 \\ - \quad 9 \quad 7 \quad .5 \quad 0 \\ \hline \$5 \quad 5 \quad 2 \quad .5 \quad 0 \end{array}$$

3. A used laptop costs \$420. The laptop was then marked up 57% before selling it. What was the sale price of the laptop?

$$420 \cdot 0.57 = \$239.40 \text{ markup}$$

$$\begin{array}{r} 4 \quad 2 \quad 0 \quad .0 \quad 0 \\ +2 \quad 3 \quad 9 \quad .4 \quad 0 \\ \hline \$6 \quad 5 \quad 9 \quad .4 \quad 0 \end{array}$$

4. The owner of a fast-food chain spends \$3.25 making a burger and then adds 115% markup to sell the burger. Calculate the selling price of the burger.

$$3.25 \cdot 1.15 = \$3.74 \text{ markup}$$

$$\begin{array}{r} 3 \quad .2 \quad 5 \\ 3 \quad .7 \quad 4 \\ \hline \$6 \quad .9 \quad 9 \end{array}$$

5. A company produces a piece of luggage for \$175.50. The piece of luggage is then mark up 68%. What is the selling price of the piece of luggage?

$$175.50 \cdot 0.68 = \$119.34 \text{ markup}$$

$$\begin{array}{r} 1 \quad 7 \quad 5 \quad .5 \quad 0 \\ +1 \quad 1 \quad 9 \quad .3 \quad 4 \\ \hline \$2 \quad 9 \quad 4 \quad .8 \quad 4 \end{array}$$



6. A clothing store sells a t-shirt for \$15.75, already marked with 20% off the original price. Calculate the original price of the t-shirt.

$$\begin{aligned}
 x - 0.20x &= 15.75 \\
 \frac{0.8x}{0.8} &= \frac{15.75}{0.8} \\
 x &= \$19.69
 \end{aligned}$$

7. A scientific calculator is 18% off and its sale price is \$29.82. What is the original price of the scientific calculator?

$$\begin{aligned}
 x - 0.18x &= 29.82 \\
 \frac{0.82x}{0.82} &= \frac{29.82}{0.82} \\
 x &= \$36.37
 \end{aligned}$$

8. The price of purchasing an Escape Room activity for a group is \$275.60. If you get it at 27% off, how much will you pay?

$$\begin{aligned}
 275.60 \cdot 0.27 &= \$74.41 \text{ off}
 \end{aligned}$$

2	7	5	.6	0
–	7	4	.4	1
\$2	0	1	.1	9

9. A store sells a computer monitor for \$645.25 after a $\frac{7}{10}$ markup. How much is the markup?

$$\begin{aligned}
 &70\% \text{ markup} \\
 x + 0.70x &= 645.25 \\
 1.7x &= 645.25 \\
 x &= 379.56 \text{ original price}
 \end{aligned}$$

6	4	5	.2	5
–3	7	9	.5	6
\$2	6	5	.6	9

10. A restaurant sells its famous bacon grilled cheese sandwich for \$4.82. The restaurant added a markup of $\frac{18}{10}$ the cost of the bacon grilled cheese sandwich. Find the cost of how much the restaurant pays for making its famous sandwich.

$$\begin{aligned}
 &180\% \text{ markup} \\
 x + 1.8x &= 4.82 \\
 \frac{2.8x}{2.8} &= \frac{4.82}{2.8} \\
 x &= \$1.72
 \end{aligned}$$



Grade 6 Accelerated
Unit 12
Lesson 5 Homework

Name Answer Key
Date _____
Period _____

For each problem, round your answer to the nearest hundredth, if necessary.

1. The original price of a wooden table is \$34. With a discount of the 25%, what is the sale price?

$$34(0.75) = \$25.50$$

2. Gary buys a pair of jeans online for \$78. The shipping company charges 8% of the item price to make the delivery, how much do the jeans and shipping cost to Gary?

$$78(1.08) = \$84.24$$

3. A customer pays \$435.89 for an Xbox Game console and a game. The customer also needs to pay 5% for taxes with his purchase. What is the total price the customer needs to pay at checkout?

$$435.89(1.05) \approx \$457.68$$

4. The sales tax rate is 4%. If Gus buys a new pair of shoes priced at \$117.93, what will be the cost including tax?

$$117.93(1.04) \approx \$122.65$$

5. A furniture company sold a sofa for \$765 and with sales tax, the sofa came out to \$815.36. What is the sales tax percentage?

$$\begin{aligned} 815.36 - 765 &= 50.36 \text{ sales tax} \\ \frac{50.36}{765} \times 100 &= 6.58\% \end{aligned}$$

6. A computer monitor cost a company \$364 and is now selling it for \$476. What is the markup percentage?

$$476 - 364 = 112 \text{ mark up}$$
$$\frac{112}{364} \times 100 = 30.77\%$$

7. Riley bought a game that cost \$45.25. After sales tax, the total is \$47.99. What is the sales tax percentage?

$$47.99 - 45.25 = 2.74 \text{ sales tax}$$
$$\frac{2.74}{45.25} \times 100 = 6.06\%$$

8. A bill for a restaurant came out to be \$137.50. The customer tips 15% of the bill. How much does the customer leave for the tip?

$$137.50(0.15) \approx \$20.63$$

9. George and his friend are celebrating each other's birthday. They decided to go to Top Golf. After appetizers and an hour of playing golf, the total bill, including the tip was \$186.55. The friends left a 20% tip. How much was the bill, excluding the tip?

$$\begin{array}{rcl} \% & 1.2 & 1 \\ \$ & \frac{186.55}{1.2} & = \frac{1}{x} \\ & 1.2x = 186.55 & \\ & x = \$155.46 & \end{array}$$

10. Some friends went to a convenience store to buy snacks to watch a home baseball game. The cost of snacks was \$87.62 and also 3% for taxes. How much was the total price of the bill?

$$87.62(1.03) \approx \$90.25$$

Grade 6 Accelerated
Unit 12
Lesson 6 Homework

Name Answer Key
Date _____
Period _____

For each problem, round your answer to the nearest hundredth, if necessary.

1. Randall bought a table priced at \$40 but is on sale for 30% off. After paying a tax of 3%, how much did Randall pay?

$$40(0.7) = 28 \quad 28(1.03) \approx \$28.84$$

Randall paid \$28.84 total for the table.

2. Amy bought a surfboard for \$175.82. Amy restored the surfboard and placed a mark up of 30% to sell in her store. A customer purchased the surfboard and paid an additional 2% in taxes. What was the total price of the surfboard?

$$175.82(1.3) = 228.566 \quad 228.566(1.02) \approx \$233.14$$

The total price of the surfboard was \$233.14.

3. A jewelry store bought a ring and mark it up 115% from the original cost of \$85.90. A customer purchased the ring and paid 8% sales tax. How much, including tax, did the customer pay for the ring?

$$85.90(2.15) = 184.685 \quad 184.685(1.08) \approx \$199.4598$$

The customer paid \$199.46 for the ring.

4. Jessica owns a furniture store with refurbished dining room tables. She purchased a table for \$275. After repairs, Jessica put the table up for sale and marked it up 100% from her original cost. After a while, the table was sold. Determine the total cost of the table including an added 11% sales tax.

$$275(2) = 550 \quad 550(1.11) \approx \$610.50$$

The total cost of the table was \$610.50.

5. Harold bought a washing machine that was marked down 60% of its original price of \$380. If the store also charged Harold 5% in taxes, how much did Harold have to pay at the cash register?

$$380(0.40) = 152 \quad 152(1.05) \approx \$159.60$$

Harold had to pay \$159.60.

6. Jimmy paid a total of \$870 for an item that had a discount of 15%. He also paid 3% in taxes after the discount was applied. Calculate how much the item cost before the discount.

$$\frac{1.03x}{1.03} = \frac{870}{1.03} \quad x = 844.66 \quad \frac{.85x}{.85} = \frac{844.66}{.85} \quad x = 993.72$$

The original price of the item was \$993.72.

7. A used car was sold for \$3,500 after taxes. The car had a markup of 110% and its new owner had to pay 8% in taxes. How much was the original price of the car?

$$\frac{3500}{1.08} = 3240.74 \quad \frac{3240.74}{2.1} = \$1543.21$$

The original price of the car was \$1543.21.

8. Rachel enters a store that was selling everything at 15% off. Rachel's total bill, before the discount, was \$137. Rachel had to pay a 2% sales tax with her purchase. How much did Rachel pay?

$$137(0.85) = 116.45 \quad 116.45(1.02) = \$118.78$$

Rachel paid \$118.78.

9. An item was sold at \$1,875 after taxes. The item had a discount of 20% and a sales tax of 10%. How much was the original price of the item?

$$\frac{1875}{1.1} = 1704.55 \quad \frac{1704.55}{0.8} = \$2130.69$$

The original price was \$2130.69.

10. Joshua owns a firewood company. Joshua bought a load of wood for \$1,780 and marked up the wood by 20%. Joshua sells the wood and charges a 3% sales tax. For how much does Joshua sell the wood?

$$1780(1.2) = 2136(1.03) = \$2200.08$$

He sells the wood for \$2200.08.

Grade 6 Accelerated
Unit 12
Lesson 7 Homework

Name Answer Key
Date _____
Period _____

1. The original value of an antique vase is \$350. That same item is now worth \$280. Find the percentage of change.

$$\frac{280-350}{350} = -\frac{70}{350} = -0.2 \rightarrow 20\% \text{ decrease}$$

2. A car was worth \$4,800. Overtime, the car has decreased 15%. How much is the car worth now?

$$0.85 \cdot 4800 = 4080$$

The car is worth \$4,080 now.

3. A company created a new box that can hold 5% more cereal. The original box held 300 grams of cereal. How much cereal does the new box hold?

$$300 \cdot 1.05 = 315$$

The new box holds 315 grams of cereal.

4. The population of a city has increased 20% over the years. A few years back the city had a population of 18,000 people. How many people does it have now?

$$18,000 \cdot 1.2 = 21,600$$

The population is 21,600.

5. A ring was bought for \$127. It has decreased in value by 18%. How much is the ring worth now?

$$0.82 \cdot 127 = 104.14$$

The ring is worth \$104.14 now.



6. A football event held over 35,750 people. Now, the event holds 26% more people. How many people are at the event?

$$35,750 \cdot 1.26 = 45,045$$

45,045 people are at the event.

7. A watch's original value was \$75. After a long time, the watch has increased in value 85%. How much is the watch worth?

$$75 \cdot 1.85 = 138.75$$

The watch is worth \$138.75 now.

8. Jorge's house had an original value was \$300,000. Now it is worth \$480,000. What is the percentage of change?

$$\frac{480,000 - 300,000}{300,000} = \frac{180,000}{300,000} = 0.6 \rightarrow 60\%$$

The house's value increased by 60%.

9. A motorcycle's original value was \$3,760. Now, the value has decreased 15%. Find the motorcycle's value now.

$$0.85 \cdot 3760 = 3196$$

The motorcycle's value is now \$3,196.

10. After improvements, the value of a car has increased 35%. The original value of the car was \$2,800. How much is the car worth now?

$$2800 \cdot 1.35 = 3780$$

The car is now worth \$3,780.



Grade 6 Accelerated
Unit 12
Lesson 8 Homework

Name Answer Key
Date _____
Period _____

1. How much money would an investment of \$3,875 in a 3-year period and 4% simple interest rate makes?
 $3875 \cdot 0.04 \cdot 3 = 465$
The investment would make \$465 in interest.

2. \$560 are put into a savings account. The account has a 6% simple interest rate. How much money would the account have by the end of 3-years?
 $560 \cdot 0.06 \cdot 3 = 100.8$ $100.8 + 560 = 660.8$
The account would have \$660.80 in it.

3. How much money would an investment of \$7,730 during a 3-year period with a 4% simple interest rate make?
 $7730 \cdot 3 \cdot 0.04 = 927.6$
The investment would make \$927.60.

4. An amount \$360 are put into a savings account. The account has a 2% simple interest rate. How much money would the account have by the end of 10 years?
 $360 \cdot 10 \cdot 0.02 = 72$
At the end of 10 years, the account would have \$432.

5. How much money would an investment of \$390 in an 3-year period with a 4% simple interest rate make?
 $390 \cdot 3 \cdot 0.04 = \46.80
The investment would make \$46.80.



6. A company needed to hire a new sales manager and expected 28 people to turn in resumes for a job interview. By the end of the process, only 17 people did. What is the percentage of error?

$$\left| \frac{28 - 17}{17} \right| = \left| \frac{11}{17} \right| = 0.647 \rightarrow 64.7\%$$

The percentage of error is 64.7%.

7. What is the percent error in using $\frac{22}{7}$ as an approximation of π ?

$$\left| \frac{\frac{22}{7} - \pi}{\pi} \right| = \left| \frac{0.001264}{\pi} \right| = 0.000402499 \rightarrow 0.04\%$$

The percent error is 0.04%.

8. An estimated 1,800 people were expected to attend a rock concert at Daily's Place, but only 1,340 attended. What is the percentage of error?

$$\left| \frac{1800 - 1340}{1340} \right| = \left| \frac{460}{1340} \right| = 0.3433 \rightarrow 34.33\%$$

The percentage of error is 34.33%.

9. Justin is preparing to run a 5K race. He wanted to run 3 miles a day but, by the end of the week, he only managed to run 2 miles per day. What is the percentage of error?

$$\left| \frac{3 - 2}{2} \right| = \left| \frac{1}{2} \right| = 0.5 = 50\%$$

The percentage of error is 50%.

10. The population of a town in the year 2000 was 38,000. It was expected to rise up to 45,000 by the end of the year 2020. But, by 2020 the population was only 36,000. What was the percentage of error?

$$\left| \frac{45,000 - 36,000}{36,000} \right| = \left| \frac{9,000}{36,000} \right| = 0.25 \rightarrow 25\%$$

The percentage of error was 25%.

